Ashritha Gugire

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RESEARCH SUMMARY

Applied Researcher specializing in machine learning, anomaly detection, and AI systems across various domains. My work covers the full research lifecycle, from designing theoretical frameworks and developing algorithms to large-scale validation and production deployment. I combine strong academic rigor with applied impact, with research contributions in leading venues such as IEEE and IISE Transactions, and a strong ability to translate complex ML insights into actionable outcomes for both technical and business stakeholders.

EDUCATION

Master of Science, Data Analytics Engineering (Spec: Machine Learning), George Mason University — ***Jan 2023 - Dec 2024***

Bachelor of Technology, Computer Science, Jawaharlal Nehru Technological University — ***Aug 2017 - June 2021***

TECHNICAL SKILLS

Machine Learning: Modelling & Detection, Neural Networks, and Natural Language Processing (NLP)
Applied Statistics: A/B Testing, Hypothesis Testing, Linear & Logistic Regression, Causal Inference, Classification
Programming: Python (Scikit-learn, Pandas, PyTorch, Keras), MATLAB, R, SQL, Bash, Unix Fundamentals
Database Platforms: SQL Server, PostgreSQL, MySQL, Snowflake, AWS (S3, RDS), Azure Databricks
Technical Leadership: Communication & collaboration, technical documentation, research presentation, code review

RESEARCH EXPERIENCE

Interpretable Anomaly Detection in Urban Mobility *(IISE Transactions)* — *Mar 2025 - Aug 2025*

- Investigated and studied multi-scale anomaly dynamics using a cross-temporal Hidden Markov Models (HMM), revealing hidden sub-daily deviations during crises and improving the reliability of anomaly detection for emergency analytics.
- Developed a bidirectional text–trajectory conversion system and a scalable feature engineering pipeline (136K records, 121 variables, 85 POI categories across 8 clusters) to transform raw mobility data into interpretable explanations, generate synthetic trajectories, and uncover 78% population-level behavioral alignment during emergencies
- Compared single-scale and multi-scale detection results and evaluated four anomaly detection strategies, reduced overall anomaly rates and achieved a 12.2% detection rate with the highest interpretability and lowest variance.

CARLA- Scalable Anomaly Detection in Urban Simulations (IARPA project) — *Aug 2025 - Nov 2025*

- Architected and proposed a contrastive representative learning framework for large-scale surveillance simulations (1M agents), designing six synthetic anomaly methods and training a triplet-loss encoder (51 to 1280D) across five anomaly profiles to jointly learn normal and anomalous behavioral representations.
- Validated CARLA as a scalable first-stage anomaly filter preceding Deloitte's OOD, Tree, and Transformer ensemble models, reducing computational load by 80% while improving throughput and recall for planted-agent detection by 40%.

Physics-Inspired Group Anomaly Detection in Human Mobility *(IEEE TKDE)* — Jan *2026 - present*

- Proposed a physics-inspired framework for group anomaly detection in human mobility that shifts from rule-based group definitions to trajectory-centric analysis, and uncovering anomalies missed by conventional co-occurrence or demographic methods.
- Developed eight trajectory representations spanning temporal, spatial, semantic, transition-based, duration-weighted, multi-scale, and group-relative views, validating their universality across real-world and synthetic datasets and linking mobility behavior to physics principles such as entropy, equilibrium, and conservation.

WORK EXPERIENCE

Center for Assurance Research and Engineering | Research Associate — *April 2024* – Present

***Industry and Government Collaboration**: Deloitte, IARPA (Intelligent Advanced Research Projects)*

- Designed advanced anomaly detection pipelines integrating graph-based techniques and clustering methods (K-Means, HDBSCAN) over 1M operational records, reducing false positives by 20% in real-time monitoring systems.
- Engineered and deployed ensemble learning models (XGBoost, LSTM, Random Forest, AdaBoost, Logistic Regression) with 18% accuracy gain, validated and hypothesis testing, for user behavior prediction in large-scale digital ecosystems.
- Investigated discrete choice modeling using Gaussian Mixture Models and ensemble methods, developing data-driven threshold methodologies through IQR analysis to address classification challenges in unlabeled mobility data.
- Led research innovation by experimenting with Transformer architectures, Autoencoders, and self-supervised learning techniques, validating findings through rigorous A/B testing across multiple datasets.
- Applied NLP and LLM-based frameworks using spaCy, BERT, GPT, LangChain, and Retrieval-Augmented Generation (RAG) pipelines for Named Entity Recognition, text summarization, and semantic context extraction from documents.
- Collaborated with multidisciplinary team of 3 researchers including domain experts in urban planning and emergency management, presented research findings and translating technical ML concepts for non-technical stakeholders

Infosys Ltd. | Data *& AI Engineering* | *India* — *Aug 2021- Dec 2022*

- Designed and optimized machine learning pipelines for enterprise systems using Python, TensorFlow, and SQL, improving prediction throughput and reducing response latency.
- Implemented predictive maintenance and fault detection systems for industrial operations, improving up time by 15% through supervised learning models and feature-based diagnostics
- Collaborated with multi-disciplinary teams to translate business objectives into deployable AI solutions, experience that informed later research in large-scale anomaly detection and MLOps automation
- Provided technical guidance to junior team members through code reviews and knowledge-sharing sessions on SQL optimization, Power BI best practices, and Python data analysis workflows.

RESEARCH VISION

Publishing, presenting, and collaborating with the brightest minds in academia and industry to refine and challenge my own thinking.

Ambition is to contribute to research so foundational, so universally significant, that it reshapes how the world understands my work and worthy of the highest recognition in science.